Soma Heritage

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Soma Heritage

Soma Heritage: Neural Network Distillation via Signal Field Resonance

Abstract

1. Introduction

1.1

1.2 Soma Heritage

1.3

2. Method

2.1

- W_Q, W_K, W_V, W_O
- $W_c \in \mathbb{R}^{n_{kv} \times k \times d_{head}}$
- $\log \gamma \in \mathbb{R}^k$

2.2

$$\mathcal{L}_{\text{total}} = w_1(t) \cdot \mathcal{L}_{\text{feature}} + w_2(t) \cdot \mathcal{L}_{\text{logit}} + w_3(t) \cdot \mathcal{L}_{\text{consistency}}$$

2.3

```
:  M_T,  M_S,  {L_1, L_2, ..., L_m}
:  M_S'

for stage = 1 to m do
    l ← L_stage
    M_S.layers[l].attention ← SignalFieldAttention()
    for i = 0 to l do
        freeze(M_S.layers[i])
    end for
    train M_S.layers[l] with three-layer loss
    freeze(M_S.layers[l])
end for
```

2.4

$$I(l) = \kappa(A_l) \cdot \|\nabla \mathcal{L}_l\|_2$$

3. Experiments

3.1

3.2

3.3 PPL

Baseline PPL	SignalField PPL			
Layer 0	22.375	23.062	**+3.07%**	
Layer 11	22.375	22.255	**-0.57%**	
Layer 23	22.375	20.011	**-10.57%**	

3.4

3.5

Baseline PPL	SignalField PPL		
WikiText-2	22.375	23.062	+3.07%
Penn Treebank	23.500	22.800	-2.98%

3.6

3.7 FLOPs

** 9SFA vs Standard Attention FLOPs**

SFA	Standard Attention		
FLOPs (seq=1024, d=512)	1.08×10^9	1.61×10^9	**-32.8%**

3.8

4. Discussion

4.1

4.2

4.3

5. Conclusion

References

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